Giuseppe Floris

PhD Student in Electronics and Computer Engineering

University of Cagliari, Italy

giuseppe.floris@unica.it | giuseppefloris.github.io
GitHub | LinkedIn | sAIfer Lab



Research profile

I am a member of sAIferLab and a PhD student in Electronics and Computer Engineering at the University of Cagliari. My research focuses on machine learning security and privacy, and on applications of machine learning to cybersecurity. In particular, I work on adversarial robustness and privacy attacks and defenses. I am interested in evaluating security and privacy risks in machine learning systems and developing methods to mitigate them.

Research experience

Visiting PhD (Supervisor: Prof. Franziska Boenisch) Nov 2025 - Apr 2026

CISPA Helmholtz Center for Information Security, Germany · SprintML Lab

Research on privacy attacks and defenses in LLMs and their tokenizers.

Research Intern Jan 2025 - Jul 2025

PLURIBUS ONE S.r.l.

Development of API-Sec, an ML-based threat detection framework for web APIs.

Member · sAIfer Lab 2024 - present

University of Cagliari, Italy

Member as a PhD Student of the sAIfer Lab at the Department of Electrical and Electronic Engineering.

Research Grant Jun 2023 - Sep 2023

University of Cagliari, Italy

Comparative analysis of robustness of image classifiers against state-of-the-art perturbation models using PyTorch.

Education

PhD in Electronics and Computer Engineering 2023 - 2026

University of Cagliari, Italy · DRIEI PhD Program

Supervisor: Prof. Battista Biggio · Co-supervisor: Dr. Maura Pintor.

MSc in Computer Engineering, Cybersecurity and Artificial Intelligence 2021 - 2023

University of Cagliari, Italy

110/110 with honors. Thesis: *Hyperparameter Optimization for Fast Minimum-norm Attacks*.
Supervisor: Prof. Battista Biggio.

BSc in Electrical, Electronic and Computer Engineering 2016 - 2020

University of Cagliari, Italy

Thesis: *Mitigation of vulnerabilities in the evolution of open source software development*.
Supervisor: Prof. Giorgio Giacinto.

Publications under review

- **Privacy attacks and DP protections in Large Language Models' tokenizers.** Attacks that infer whether a website was used to train an LLM tokenizer, and differentially private tokenizer training with website-level privacy guarantees.
- **Benchmarking privacy risk on LLMs.** A statistical test for distribution shift in LLM privacy benchmarks, and a controlled benchmark using canaries to compare membership inference attacks when natural text provides too little signal.

Publications

ModSec-AdvLearn: Countering Adversarial SQL Injections With Robust Machine Learning

Giuseppe Floris, Christian Scano, Biagio Montaruli, Luca Demetrio, Andrea Valenza, Luca Compagna, Davide Ariu, Luca Piras, Davide Balzarotti, Battista Biggio.

IEEE Transactions on Information Forensics and Security (TIFS), vol. 20, pp. 6693-6705, 2025.

HO-FMN: Hyperparameter Optimization for Fast Minimum-Norm Attacks

Raffaele Mura, **Giuseppe Floris**, Luca Scionis, Giorgio Piras, Maura Pintor, Ambra Demontis, Giorgio Giacinto, Battista Biggio.

Neurocomputing, vol. 616, article 128918, 2025.

ModSec-Learn: Boosting ModSecurity with Machine Learning

Christian Scano, **Giuseppe Floris**, Biagio Montaruli, Luca Demetrio, Andrea Valenza, Luca Compagna, Davide Ariu, Luca Piras, Davide Balzarotti, Battista Biggio.

21st International Conference on Distributed Computing and Artificial Intelligence (DCAI), Salamanca, Spain, 26-28 June 2024.

Improving Fast Minimum-Norm Attacks with Hyperparameter Optimization

Giuseppe Floris, Raffaele Mura, Luca Scionis, Giorgio Piras, Maura Pintor, Ambra Demontis, Battista Biggio.

31st European Symposium on Artificial Neural Networks, Computational Intelligence and Machine Learning (ESANN), Bruges, Belgium, 4-6 October 2023.

Technical skills

Programming: Python (AI/ML and LLM libraries).

Machine learning & AI security: applications to cybersecurity, hyperparameter optimization, adversarial attacks and robustness, privacy attacks, differential privacy.

Languages

Italian: native **English:** B2

Additional information

Citizenship: Italian.